



配分函数与宏观量

 $\alpha(T, \mu) \quad p(T)$

 Boltzmann 分布: $n_i = g_i e^{-\alpha - \beta \epsilon_i}$

$$N = \sum_i n_i = \sum_i g_i e^{-\alpha - \beta \epsilon_i} = e^{-\alpha} \sum_i g_i e^{-\beta \epsilon_i} = e^{-\alpha} Z$$

$$e^{-\alpha} = \frac{N}{Z} \Leftrightarrow \alpha = \ln \frac{Z}{N} \leftarrow \text{配分函数 } Z = Z(\beta, \mu)$$

$$\begin{aligned} \bar{E} &= \sum_i n_i \epsilon_i = \sum_i \frac{N}{Z} g_i e^{-\beta \epsilon_i} \epsilon_i = \frac{N}{Z} \sum_i g_i e^{-\beta \epsilon_i} \epsilon_i \\ &= \frac{N}{Z} \frac{\partial Z}{\partial (-\beta)} = - \frac{N}{Z} \frac{\partial Z}{\partial \beta} \\ &= -N \frac{\partial (\ln Z)}{\partial \beta} \end{aligned}$$

 热力学第一定律 $d\bar{E} = \delta W + \delta Q$

$$d\bar{E} = d\left(\sum_i n_i \epsilon_i\right) = \sum_i n_i d\epsilon_i + \sum_i \epsilon_i dn_i$$

$$\delta W = \vec{Y} d\vec{y} \leftarrow \text{广义位移}$$

↑
广义力

$$\epsilon_i(\vec{y}) \Rightarrow \vec{Y} d\vec{y} = \sum_i n_i d\epsilon_i$$

$$\sum_i Y_k dy_k = \sum_i n_i \sum_k \frac{\partial \epsilon_i}{\partial y_k} dy_k = \sum_k \left(\sum_i n_i \frac{\partial \epsilon_i}{\partial y_k} \right) dy_k$$

↑
 Y_k

$$\begin{aligned} Y_k &= \sum_i n_i \frac{\partial \epsilon_i}{\partial y_k} = \sum_i g_i e^{-\alpha} e^{-\beta \epsilon_i} \frac{\partial \epsilon_i}{\partial y_k} = \frac{N}{Z} \sum_i g_i \cdot \frac{1}{-\beta} \cdot \frac{\partial (e^{-\beta \epsilon_i})}{\partial y_k} \\ &= - \frac{N}{Z} \cdot \frac{1}{\beta} \frac{\partial Z}{\partial y_k} = - \frac{N}{\beta} \frac{\partial \ln Z}{\partial y_k} \end{aligned}$$

$$\text{取 } y_k = V; \quad - \frac{N}{\beta} \frac{\partial \ln Z}{\partial V} = \delta W = -Y_k \partial y_k = -Y_k \partial V \Rightarrow Y_k = -P$$

$$\delta Q = d\bar{E} - \sum_k Y_k dy_k$$

$$= -N d\left(\frac{\partial \ln Z}{\partial \beta}\right) + \frac{N}{\beta} \sum_k \frac{\partial \ln Z}{\partial y_k} dy_k$$



$$z(\beta, y) \Rightarrow d(\ln z) = \frac{\partial \ln z}{\partial \beta} d\beta + \sum_i \frac{\partial \ln z}{\partial y_i} dy_i$$

$$\begin{aligned}\delta Q &= -N d\left(\frac{\partial \ln z}{\partial \beta}\right) + \frac{N}{\beta} \left[d(\ln z) - \left(\frac{\partial \ln z}{\partial \beta}\right) d\beta \right] \\ &= \frac{N}{\beta} \left(d(\ln z) - \left(\frac{\partial \ln z}{\partial \beta}\right) d\beta - \beta d\left(\frac{\partial \ln z}{\partial \beta}\right) \right) \\ &= \frac{N}{\beta} d\left[\ln z - \beta \frac{\partial \ln z}{\partial \beta} \right]\end{aligned}$$

因为 β 和 $\frac{1}{\beta}$ 都是 δQ 的积分因子, 所以 $\beta = \frac{1}{kT}$ (设 $k = k_B$)

$$\delta Q = T dS$$

$$dS = Nk d\left[\ln z - \beta \frac{\partial \ln z}{\partial \beta} \right]$$

$$S = Nk \left[\ln z - \beta \frac{\partial \ln z}{\partial \beta} \right] + S'$$

對於半經典統計, $S' = Nk \ln \frac{e}{N}$

$$S = Nk \left[\ln \frac{e^2}{N} - \beta \frac{\partial \ln z}{\partial \beta} \right]$$

$$\text{自由能 } F = \bar{E} - TS$$

$$= -N \frac{\partial \ln z}{\partial \beta} - NkT \left[\ln \frac{e^2}{N} - \beta \frac{\partial \ln z}{\partial \beta} \right] = -NkT \ln \frac{e^2}{N}$$

$$\begin{aligned}\text{化學勢: } \mu &= \left(\frac{\partial F}{\partial N} \right)_{T, V} = \cancel{-kT \ln \frac{e^2}{N}} = \cancel{-kT \left(\ln \frac{e^2}{N} - \right)} \\ &= -kT \ln \frac{e^2}{N} \Rightarrow \alpha = \ln \frac{e^2}{N} = -\frac{\mu}{kT}\end{aligned}$$

$$NkT \ln \frac{N}{e^2}$$

$$\begin{aligned}&+ kT (N \ln N - N \ln e^2) \\ &\quad \downarrow \\ &\ln N + 1 - \ln e^2\end{aligned}$$



量子统计

2021.12.03

半经典: $W_s\{n_i\} = \prod_i \frac{g_i^{n_i}}{n_i!}$ 最可几分布: $n_i = g_i e^{-\alpha - \beta \epsilon_i}$ (平衡)

$$W_{sm}\{n_i\} = \prod_i \frac{g_i^{n_i}}{n_i!} \Big|_{n_i = g_i \exp(-\alpha - \beta \epsilon_i)}$$

$$\ln W_{sm}\{n_i\} \approx \sum_i [n_i \ln g_i - n_i (\ln n_i - 1)]$$

$$= \sum_i n_i \left(\ln \frac{g_i}{n_i} - 1 \right)$$

$$= \sum_i n_i (\alpha + \beta \epsilon_i - 1)$$

$$\text{回忆 } z = \sum_i g_i e^{-\beta \epsilon_i}$$

$$= \frac{1}{N} (\alpha + 1) N + \beta \bar{E}$$

$$\alpha = \ln \frac{z}{N}$$

$$= (\ln \frac{z}{N} + 1) N - N \beta \frac{\partial \ln z}{\partial \beta}$$

$$\bar{E} = -N \frac{\partial \ln z}{\partial \beta}$$

$$= N \left(\ln z - \beta \frac{\partial \ln z}{\partial \beta} \right) + N (1 - \ln N)$$

$$= N \left(\ln \frac{e z}{N} - \beta \frac{\partial \ln z}{\partial \beta} \right)$$

$$= \frac{S}{k} \quad [S' = N k \ln \frac{e}{N}]$$

$$\Rightarrow S = k \ln W_{sm}\{n_i\}$$

定域粒子: $W_s\{n_i\} = N! \prod_i \frac{g_i^{n_i}}{n_i!}$, $S' = 0$

$$S = k N \left(\ln z - \beta \frac{\partial \ln z}{\partial \beta} \right), \quad F = -N k T \ln z$$



粒子状态的半经典描述

(1) 当 $\frac{\Delta E_i}{kT} \ll 1$, 粒子能级半连续

(2) 当 $\lambda = \frac{h}{p}$, $h \ll p$, 粒子波动性弱

$$H = H(\vec{q}, \vec{p}), \quad r$$

$$\text{正则运动} \quad \frac{dq_i}{dt} = \frac{\partial H}{\partial p_i}, \quad \frac{dp_i}{dt} = -\frac{\partial H}{\partial q_i}$$

$$\text{若 } H \text{ 不显含时间, } H(\vec{q}, \vec{p}) = E, \quad \Delta q_i, \Delta p_i \approx h$$

在能量半连续条件下, 对于自由度为 r 的粒子的量子数足够大

— 量子态的相体积是 h^r

$$\text{— 维谐振子: } \Omega(E) = \frac{E}{\nu}, \quad \Delta \Omega(E_n) = 1$$

半连续: $\frac{\Delta E_i}{kT} \ll 1$, 如 $m \sim 10^{-27} \text{ kg}$, $L \sim 10^{-2} \text{ m}$, $T \sim 300 \text{ K}$

$$\frac{\Delta E_i}{kT} \leq 10^{-5}$$

$$g_i \sim g(E) dE = \frac{1}{h^r} \frac{d\Omega(E)}{dE} dE \sim \frac{\Delta \Omega(E_i)}{h^r \Delta E_i} \Delta E_i$$

$$n_i \sim n(E) dE$$

$$Z(\beta, \gamma) = \sum_i g_i e^{-\beta E_i} \sim \int g(E) e^{-\beta E} dE$$

$$\text{理想气体} \quad g(E) = \frac{2\pi V (2m)^{\frac{3}{2}}}{h^3} \sqrt{E}$$

$$Z(\beta, \gamma) = \int_0^\infty g(E) e^{-\beta E} dE$$

$$= \frac{2\pi V (2m)^{\frac{3}{2}}}{h^3} \int_0^\infty e^{-\beta E} E^{\frac{1}{2}} dE$$

$$= \frac{2\pi V (2m)^{\frac{3}{2}}}{\pi^{\frac{1}{2}} h^3} \cdot \Gamma\left(\frac{3}{2}\right) = \frac{2\pi V (2m)^{\frac{3}{2}}}{\pi^{\frac{1}{2}} h^3} \cdot \frac{1}{2} \sqrt{\pi} = \frac{(2\pi m)^{\frac{3}{2}} V}{\pi^{\frac{1}{2}} h^3} = \frac{V}{h^3} \left(\frac{2\pi m}{\beta}\right)^{\frac{3}{2}}$$



$$e^{\alpha} = \ln \frac{Z}{N} = \ln \frac{V}{N} \left(\frac{2\pi m}{h^2 \beta} \right)^{\frac{3}{2}}$$

$$\Rightarrow e^{\alpha} = \frac{V}{N} \left(\frac{2\pi m}{h^2 \beta} \right)^{\frac{3}{2}} = \frac{V}{N} \left(\frac{2\pi m kT}{h^2} \right)^{\frac{3}{2}}$$

因为 $e^{\alpha} \gg 1$, 所以 $\frac{N}{V} \ll 1$ (稀薄), T 高 (高温)

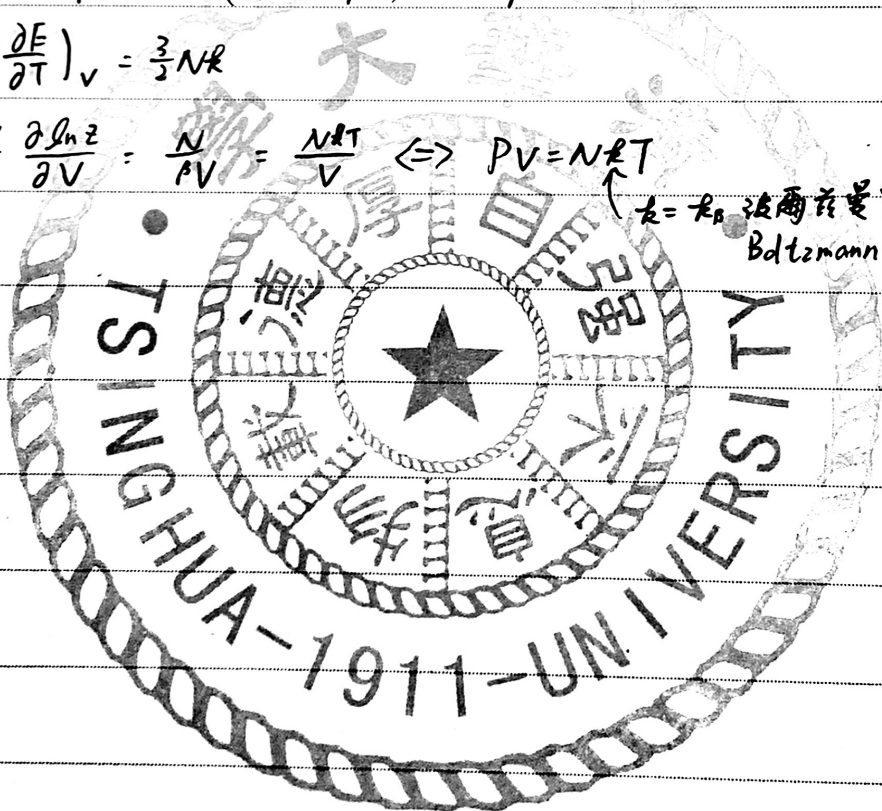
m 大 (重质量)

$$\bar{E} = -N \frac{\partial \ln Z}{\partial \beta} = -N \left(\frac{3}{2} \cdot \left(-\frac{1}{\beta} \right) \right) = \frac{3}{2} \frac{N}{\beta} = \frac{3}{2} NkT$$

$$C_V = \left(\frac{\partial \bar{E}}{\partial T} \right)_V = \frac{3}{2} Nk$$

$$P = \frac{N}{\beta} \frac{\partial \ln Z}{\partial V} = \frac{N}{\beta V} = \frac{NkT}{V} \Leftrightarrow PV = NkT$$

$k = k_B$ 玻尔兹曼常数
Boltzmann Constant





$$\theta^v = \frac{h\nu}{k} \Rightarrow \rho h\nu = \rho k \theta^v = \frac{\theta^v}{T} = \kappa(T) \quad \frac{dT}{dx} = -\frac{\theta^v}{x^2}$$

$$C_v^v = \left(\frac{\partial \bar{E}^v}{\partial T} \right)_v = N h\nu \frac{-e^{-\kappa}}{(1-e^{-\kappa})^2} \cdot \left(-\frac{x^2}{\theta^v} \right) = N k \cdot \frac{e^{-x} x^2}{(e^x - 1)^2}$$

原子内电子热容: $\theta^v = \frac{\Delta E^e}{k} \sim 10^4 \sim 10^5 K \gg T$ (常温)

Maxwell 速度分布律

$$E_i = \epsilon^x + \epsilon^y + \epsilon^z \quad g_i = g_a^x \left(\frac{1}{h^3} dx dy dz dp_x dp_y dp_z \right)$$

$$n = g_i e^{-\alpha - \beta E_i} \Rightarrow n(p_x, p_y, p_z) dp_x dp_y dp_z = \left[\sum_a g_a^x e^{-\alpha - \beta E_a^x} \right]$$

$$\frac{1}{h^3} e^{-\alpha - \frac{\beta}{2m} (p_x^2 + p_y^2 + p_z^2)} dp_x dp_y dp_z \iint e^{-\beta U_a} dx dy dz$$

$$N \iiint n(p_x, p_y, p_z) dp_x dp_y dp_z$$

$$n(p_x, p_y, p_z) = N \cdot \frac{\exp\left(-\frac{\beta}{2m} (p_x^2 + p_y^2 + p_z^2)\right)}{\iiint \exp\left(-\frac{\beta}{2m} (p_x^2 + p_y^2 + p_z^2)\right) dp_x dp_y dp_z}$$

$$= \frac{\left(\frac{2\pi m}{\beta}\right)^{\frac{3}{2}}}{\left(\frac{2\pi m}{\beta}\right)^{\frac{3}{2}} \cdot (\pi)^{\frac{3}{2}}} \exp\left(-\frac{1}{2m\beta} (p_x^2 + p_y^2 + p_z^2)\right)$$

$$n(v_x, v_y, v_z) = N \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} \exp\left(-\frac{m}{2kT} (v_x^2 + v_y^2 + v_z^2)\right)$$

$$\bar{v}_x = \bar{v}_y = \bar{v}_z = 0$$

$$n(v) dv = N \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} \exp\left(-\frac{mv^2}{2kT}\right) \cdot 4\pi v^2 dv$$

$$\bar{v}^2 = 4\pi \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} \int_0^\infty \exp\left(-\frac{mv^2}{2kT}\right) v^4 dv = \int_0^\infty \exp(-x) \cdot \left(\frac{2kT}{m}\right)^{\frac{3}{2}} x^{\frac{5}{2}} \cdot \frac{\sqrt{2kT}}{2} \frac{1}{\sqrt{x}} dx$$

$$x = \frac{mv^2}{2kT} \quad v = \sqrt{\frac{2kT}{m}} x = 4\pi \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} \cdot \left(\frac{2kT}{m}\right)^{\frac{5}{2}} \cdot \Gamma\left(\frac{5}{2}\right)$$

$$= \frac{4}{\pi} \cdot \frac{1}{\sqrt{\pi}} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi} = \frac{4}{\pi}$$



量子统计

2021.12.10

设固体有 N 个原子，由固体的自由度是 $3N$

$$E = \sum_i \frac{1}{2} m \dot{x}_i^2 + \sum_{i,j} \frac{1}{2} C_{ij} x_i x_j$$

$\downarrow$ 平动 $\downarrow$ 转动 $\downarrow$ 3N-1 自由度

把 $\vec{q} = Q \vec{x}$, $Q^T Q = I$

$$E = \sum_i \frac{1}{2} m v_i^2 = \frac{1}{2} \sum_i (2\pi \nu_i)^2 q_i^2$$

Einstein 假设 (假设原子频率一致), $E = (n + \frac{1}{2}) h \nu$
 $= (n + \frac{1}{2}) h \nu$

$$Z(\nu) = Z(\rho) = \sum_{n=0}^{\infty} e^{-\rho(n+\frac{1}{2})h\nu} = e^{-\frac{\rho h \nu}{2}} \sum_{n=0}^{\infty} e^{-\rho n h \nu} = e^{-\frac{\rho h \nu}{2}} \frac{1}{1 - e^{-\rho h \nu}}$$

$$\bar{E} = 3N \left(\frac{1}{2} h \nu + \frac{h \nu}{e^{\rho h \nu} - 1} \right)$$

$$C_V = \left(\frac{\partial E}{\partial T} \right)_V = 3N k \epsilon(x), \quad x = \frac{\theta_E}{T}, \quad \epsilon(x) = \frac{x^2 e^x}{(e^x - 1)^2}$$

当 $T \gg \theta_E$, $x \ll 1$, $\epsilon(x) \rightarrow 1$, $C_V \approx 3Nk$
 当 $T \ll \theta_E$, $x \gg 1$, $\epsilon(x) \approx x^2 e^{-x}$, $C_V = 3Nk \left(\frac{\theta_E}{T} \right)^2 e^{-\frac{\theta_E}{T}}$

顺磁物质的磁性: $\vec{\mu} = \frac{e \hbar}{2m_e} g \vec{J} = \mu_B g \vec{J}$

$$\begin{cases} \vec{\mu}_L = g_L \frac{\mu_B}{\hbar} \vec{L}, & g_L = 1 \\ \vec{\mu}_S = g_S \frac{\mu_B}{\hbar} \vec{S}, & g_S = 2 \end{cases}$$

Lande, $g = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}$

$$W_{Ze} = -\vec{\mu} \cdot \vec{B} = -\mu_B \vec{\mu} \cdot \vec{H} = -\mu_B \mu_B g m_J H$$



$$n(m_j) = (2J+1) \exp(-\alpha - \beta \epsilon_f) \\ - \mu_0 \mu_B g m_j H / kT$$

$$= n_0 \exp(\mu_0 \mu_B g m_j H / kT)$$

$$n = \sum_{m_j=-J}^J n(m_j) = n_0 \sum_{m_j=-J}^J \exp(\mu_0 \mu_B g m_j H / kT)$$

$$= n_0 Z(H, T)$$

$$\text{磁化强度 } m = \sum_{m_j} \mu_0 g m_j n(m_j)$$

$$= \sum_{m_j} \mu_0 g m_j \cdot \frac{n}{2} \exp(\mu_0 \mu_B g m_j H / kT)$$

$$= \frac{n}{2} kT \frac{\partial \ln Z}{\partial (\mu_0 H)} = n kT \frac{\partial \ln Z}{\partial (\mu H)}$$

$$Z = \exp\left(-\frac{\mu_0 \mu_B g J H}{kT}\right) \cdot \frac{1 - \exp\left(\frac{\mu_0 \mu_B g (2J+1) H}{kT}\right)}{1 - \exp\left(\frac{\mu_0 \mu_B g H}{kT}\right)}$$

$$= \frac{\sinh\left(\frac{\mu_0 \mu_B g (J+\frac{1}{2}) H}{kT}\right)}{\sinh\left(\frac{\mu_0 \mu_B g \frac{1}{2} H}{kT}\right)}$$

$$m = n kT \coth\left(\frac{\mu_0 \mu_B g (J+\frac{1}{2}) H}{kT}\right) \cdot \frac{\mu_0 g (J+\frac{1}{2})}{kT}$$

$$= n kT \coth\left(\frac{\mu_0 \mu_B g \frac{1}{2} H}{kT}\right) \cdot \frac{\mu_0 g \frac{1}{2}}{kT}$$



雙原子分子熱容量

$$\varepsilon = \varepsilon_{\text{平}}^{\text{平}} + \varepsilon_{\text{分}}^{\text{分}} + \varepsilon_{\text{振}}^{\text{振}} + \varepsilon_{\text{轉}}^{\text{轉}}$$

平動 分動 振動 轉動

$$g_i = g_{\text{平}}^{\text{平}} g_{\text{分}}^{\text{分}} g_{\text{振}}^{\text{振}} g_{\text{轉}}^{\text{轉}}$$

$$z(\beta, V) = z_{\text{平}}(\beta, V) + z_{\text{分}}(\beta) z_{\text{振}}(\beta) z_{\text{轉}}(\beta)$$

$$\bar{E} = -N \frac{\partial \ln z}{\partial \beta} = \frac{3}{2} N k T$$

$$C_V = C_V^{\text{平}} + C_V^{\text{分}} + C_V^{\text{振}} + C_V^{\text{轉}} \rightarrow C_V^{\text{平}} = \frac{3}{2} N k$$

轉動熱容量, $\hat{H} = \frac{\hat{L}^2}{2I} \Rightarrow \varepsilon_l = \frac{l(l+1)}{2I} \hbar^2, g_l = (2l+1)$

$$\Omega(\varepsilon^{\text{轉}}) = 8\pi^2 I \varepsilon, g(\varepsilon) = \frac{8\pi^2 I}{\hbar^2}$$

$$z^{\text{轉}}(\beta, V) = \int_0^\infty e^{-\beta \varepsilon^{\text{轉}}} \frac{8\pi^2 I}{\hbar^2} d\varepsilon^{\text{轉}} = \frac{8\pi^2 I}{\beta \hbar^2}$$

$$\bar{E}^{\text{轉}}(T) = -N \frac{\partial \ln z^{\text{轉}}}{\partial \beta} = N \cdot \frac{1}{\beta} = N k T$$

特徵溫度 $\theta^{\text{轉}} = \frac{\Delta \varepsilon^{\text{轉}}}{k}, \theta^{\text{轉}} = \frac{\hbar^2}{8\pi^2 I k}$ [低溫時不適用]

$$z^{\text{轉}}(T) = \sum_{l=0}^{\infty} (2l+1) e^{-\beta \cdot \frac{l(l+1)\hbar^2}{2I}} = \sum_{l=0}^{\infty} (2l+1) e^{-\frac{l(l+1)\theta^{\text{轉}}}{T}}$$

$$\approx \int_0^\infty (2l+1) e^{-\frac{\theta^{\text{轉}}}{T} l(l+1)} dl = \frac{T}{\theta^{\text{轉}}} = \frac{8\pi^2 I k T}{\hbar^2} = \frac{8\pi^2 I \rho}{\hbar^2}$$

振動熱容量 $\mu = \frac{m_1 m_2}{m_1 + m_2}$ 的線性振子。

$$\varepsilon_v = (n + \frac{1}{2}) \hbar \nu, g_v = 1$$

$$z_v(\beta) = \sum_n e^{-\beta (n + \frac{1}{2}) \hbar \nu} = e^{-\frac{\beta}{2} \hbar \nu} \sum_n e^{-\beta n \hbar \nu} = \frac{e^{-\frac{\beta}{2} \hbar \nu}}{1 - e^{-\beta \hbar \nu}} = \frac{2}{\sinh(\frac{\beta \hbar \nu}{2})}$$

$$\bar{E}_v = -N \ln \frac{\partial z_v}{\partial \beta} = N \left(\frac{1}{2} \hbar \nu + \frac{\hbar \nu}{1 - e^{-\beta \hbar \nu}} \right) = N \hbar \nu \left(\frac{1}{2} + \frac{1}{1 - e^{-\beta \hbar \nu}} \right)$$



量子統計

2021. 12. 08

$$z = \sum_i g_i e^{-\beta \epsilon_i} \rightarrow \int e^{-\beta \epsilon} g(\epsilon) d\epsilon$$

氣體單原子運動 (容器內) $\Omega(\epsilon) = V \cdot \frac{4}{3}\pi (2m\epsilon)^{\frac{3}{2}}$

$$g(\epsilon) = \frac{d\Omega(\epsilon)}{h^3 d\epsilon} = \frac{2\pi V (2m)^{\frac{3}{2}}}{h^3} \sqrt{\epsilon}$$

$$z(\beta, V) = \int_0^\infty e^{-\beta \epsilon} \frac{2\pi V (2m)^{\frac{3}{2}}}{h^3} \sqrt{\epsilon} d\epsilon = \frac{2\pi V (2m)^{\frac{3}{2}}}{h^3 \beta^{\frac{3}{2}}} \cdot \frac{1}{2} \sqrt{\pi} = \frac{V}{h^3} \left(\frac{2\pi m}{\beta} \right)^{\frac{3}{2}}$$

$$e^{-\alpha} = \frac{z}{N} = \frac{V}{N} \left(\frac{2\pi m kT}{h^2} \right)^{\frac{3}{2}}$$

$$\bar{E} = -N \frac{\partial \ln z}{\partial \beta} = N \cdot \frac{3}{2} \cdot \frac{1}{\beta} = \frac{3}{2} N kT$$

$$C_V = \left(\frac{\partial \bar{E}}{\partial T} \right)_V = \frac{3}{2} N k, \quad P = \frac{N}{\beta} \frac{\partial \ln z}{\partial V} = \frac{N kT}{V} \Rightarrow PV = N kT$$

$$F(T, V) = -N kT \ln \frac{z}{N} = -N kT (\ln z - \ln N + 1) = -N kT \left(\ln \frac{V}{N} + \frac{3}{2} \ln kT + \frac{1}{2} \ln \frac{2\pi m}{h^2} + 1 \right)$$

$$S = -\left(\frac{\partial F}{\partial T} \right)_V = N k \left(\frac{1}{2} \ln kT + \ln \frac{V}{N} + 1 + \frac{5}{2} \right)$$

$$\mu(T, \frac{N}{V}) = \left(\frac{\partial F}{\partial N} \right)_{T, V} = kT \left(\ln \frac{N}{V} - \frac{3}{2} \ln kT - 1 \right)$$

雙原子分子理想氣體
(不考慮耦合)

$$\epsilon_i = \epsilon_i^t + \epsilon_i^r$$

↑ ↑
平動 內秉能量

簡並度: $g_i^t g_i^r$

$$z(\beta, V) = \sum_i e^{-\beta \epsilon_i} g_i = \sum_j \sum_k e^{-\beta (\epsilon_j^t + \epsilon_k^r)} g_j^t g_k^r = z^t(\beta, V) z^r(\beta)$$

$$P = \frac{N}{\beta} \frac{\partial \ln z}{\partial V} = \frac{N}{\beta} \frac{\partial \ln z^t}{\partial V} = \frac{N kT}{V} \text{ 不變}$$

$$\bar{E} = -N \frac{\partial \ln z}{\partial \beta} = -N \left(\frac{\partial \ln z^t}{\partial \beta} + \frac{\partial \ln z^r}{\partial \beta} \right) = -N \left(-\frac{3}{2} kT + \frac{\partial \ln z^r}{\partial \beta} \right) = \frac{3}{2} N kT - N \frac{\partial \ln z^r(\beta)}{\partial \beta}$$



量子统计

2021.12.13

2. (7.1) 这是三维无限深势阱的能级. ~~$V = (\frac{L}{2})^3$~~ $V = (\frac{L}{2})^3$

$$\epsilon_{n_x n_y n_z} = \frac{1}{2m} (2\pi\hbar)^2 (n_x^2 + n_y^2 + n_z^2) \cdot L^{-2}$$

$$= \frac{1}{2m} (\pi\hbar)^2 (n_x^2 + n_y^2 + n_z^2) \cdot V^{-\frac{2}{3}}$$

$$P = - \sum_{n_x} \sum_{n_y} \sum_{n_z} a_{n_x n_y n_z} \frac{(\pi\hbar)^2}{2m} (n_x^2 + n_y^2 + n_z^2) \cdot \left(-\frac{2}{3}\right) V^{-\frac{5}{3}}$$

$$= \frac{2}{3} \cdot V^{-1} \sum_{n_x} \sum_{n_y} \sum_{n_z} a_{n_x n_y n_z} \frac{(\pi\hbar)^2}{2m} (n_x^2 + n_y^2 + n_z^2) V^{-\frac{2}{3}}$$

$$= \frac{2U}{3V}$$

$$3. (7.3) \quad z_1^* = \sum_i g_i e^{-\beta \epsilon_i^*} = \sum_i g_i e^{-\beta(\epsilon_i + \Delta)} = e^{-\beta \Delta} z_1$$

$$\bar{E}^* = - \frac{N}{z_1^*} \frac{\partial z_1^*}{\partial \beta} = -N \frac{\partial}{\partial \beta} (\ln z_1 - \beta \Delta) = \bar{E} + N \Delta$$

$$P^* = \frac{N}{\beta} \frac{\partial \ln z_1^*}{\partial V} = \frac{N}{\beta} \frac{\partial}{\partial V} (\ln z_1 - \beta \Delta) = P$$

$$S^* = Nk \left[\ln z_1^* - \beta \frac{\partial \ln z_1^*}{\partial \beta} \right] + S' = Nk \left[\ln z_1 - \beta \Delta + \beta \Delta - \beta \frac{\partial \ln z_1}{\partial \beta} \right] + S' = S$$

$$F^* = \bar{E}^* - T S^* = \bar{E} + N \Delta - T S = F + N \Delta$$

$$\mu^* = \left(\frac{\partial F}{\partial N} \right)_{T,V} + \left(\frac{\partial (N \Delta)}{\partial N} \right)_{T,V} = \mu + \Delta$$

$$C_V^* = \left(\frac{\partial \bar{E}^*}{\partial T} \right) = \left(\frac{\partial \bar{E}}{\partial T} \right) = C_V$$

$$5) = 4. (7.11) \quad n(p_x, p_y) = \frac{N}{\iint \exp \left[-\frac{L}{2m} (p_x^2 + p_y^2) \right] dp_x dp_y} = N \frac{\exp \left[-\frac{L}{2m} (p_x^2 + p_y^2) \right]}{\frac{2m}{\rho} \cdot \pi} dp_x dp_y$$

$$\Rightarrow n(v_x, v_y) = N \cdot \frac{m}{2\pi kT} \cdot \exp \left[-\frac{m}{2kT} (v_x^2 + v_y^2) \right]$$



$$n(v)dv = N \cdot \frac{m}{2\pi kT} \exp\left(-\frac{mv^2}{2kT}\right) \cdot 2\pi v dv$$

$$\bar{v} = \int_0^\infty v p(v) dv = \int_0^\infty v \cdot \frac{n(v)}{N} dv = \int_0^\infty \frac{m}{2kT} \exp\left(-\frac{mv^2}{2kT}\right) 2\pi v^2 dv$$

$$\left[\begin{aligned} t &= \frac{mv^2}{2kT}, \quad v = \sqrt{\frac{2kT}{m}} t \\ dv &= \frac{1}{2} \sqrt{\frac{2kT}{m}} t^{-\frac{1}{2}} dt \end{aligned} \right] = \int_0^\infty t \exp(-t) \cdot 2 \cdot \frac{1}{2} \sqrt{\frac{2kT}{m}} t^{-\frac{1}{2}} dt$$
$$= \sqrt{\frac{2kT}{m}} \Gamma\left(\frac{3}{2}\right) = \sqrt{\frac{2kT}{m}} \cdot \frac{1}{2} \sqrt{\pi}$$
$$= \sqrt{\frac{\pi kT}{2m}}$$

$$\frac{d \ln p(v)}{dv} = \frac{d}{dv} \left(-\frac{mv^2}{2kT} + \ln v \right)$$
$$= -\frac{m}{kT} v + \frac{1}{v}$$

$$\left. \frac{d \ln p(v)}{dv} \right|_{v=v_m} = -\frac{mv_m}{kT} + \frac{1}{v_m} = 0 \Leftrightarrow v_m = \sqrt{\frac{kT}{m}}$$

$$v_s^2 = \int_0^\infty v^2 p(v) dv = \int_0^\infty \frac{m}{2kT} \exp\left(-\frac{mv^2}{2kT}\right) \cdot 2\pi v \cdot v^2 dv$$
$$= \int_0^\infty \frac{mv^2}{2kT} \exp\left(-\frac{mv^2}{2kT}\right) \cdot dv^2$$
$$= \int_0^\infty t \exp(-t) \cdot \frac{2kT}{m} dt$$
$$= \frac{2kT}{m} \Gamma(2) = \frac{2kT}{m} \cdot 1! = \frac{2kT}{m}$$

$$v_s = \sqrt{\frac{2kT}{m}}$$

$$6. (7.18) \quad z_v(\beta) = \frac{e^{-\frac{\beta}{2} \hbar v}}{1 - e^{-\beta \hbar v}} = \frac{2}{\sinh\left(\frac{\beta}{2} \hbar v\right)} = \frac{2}{\sinh\left(\frac{1}{2} \frac{\theta_v}{T}\right)}$$

$$S^v = Nk \left[\ln z_v - \beta \frac{\partial \ln z_v}{\partial \beta} \right]$$

$$= Nk \left[\ln 2 - \ln \sinh\left(\frac{1}{2} \frac{\theta_v}{T}\right) + \beta \coth\left(\frac{\beta}{2} \hbar v\right) \cdot \frac{1}{2} \hbar v \right]$$

$$= Nk \left[\frac{1}{2} \beta \hbar v - \ln(1 - e^{-\beta \hbar v}) + \frac{1}{2} \beta \hbar v \cdot \frac{e^{\beta \hbar v} + 1}{e^{\beta \hbar v} - 1} \right]$$



$$\begin{aligned} S^V &= Nk \left[-\frac{1}{2} \ln(1 - e^{-\beta h\nu}) + \frac{1}{2} \beta h\nu + \frac{\beta h\nu}{e^{\beta h\nu} - 1} \right] \\ &= Nk \left[\frac{\beta h\nu}{e^{\beta h\nu} - 1} - \ln(1 - e^{-\beta h\nu}) \right] \\ &= Nk \left[\left(\frac{\theta_v}{T} \right) \cdot \frac{1}{e^{\frac{\theta_v}{T}} - 1} - \ln(1 - e^{-\frac{\theta_v}{T}}) \right] \end{aligned}$$

例 (7.2) 定域系统满足半经典分布.

$$n_l = g_l \exp(-\alpha - \beta \epsilon_l) = \exp(-\alpha - \beta \epsilon_l), \quad l=1, 2.$$

$$Z = \exp(-\beta \epsilon_1) + \exp(-\beta \epsilon_2) = \exp(-\beta \epsilon_1) [1 + \exp(-\beta(\epsilon_2 - \epsilon_1))]$$

$$\begin{aligned} \bar{E} &= -N \frac{\partial \ln Z}{\partial \beta} = -N \left(-\epsilon_1 + \frac{\exp(-\beta(\epsilon_2 - \epsilon_1))}{1 + \exp(-\beta(\epsilon_2 - \epsilon_1))} \cdot (-(\epsilon_2 - \epsilon_1)) \right) \\ &= N \left[\epsilon_1 + \frac{\epsilon_2 - \epsilon_1}{\exp[\beta(\epsilon_2 - \epsilon_1)] + 1} \right] \end{aligned}$$

$$\begin{aligned} S &= Nk \left[\ln Z - \beta \frac{\partial \ln Z}{\partial \beta} \right] \\ &= Nk \ln Z + \beta k \bar{E} = Nk \left[-\beta \epsilon_1 + \ln(1 + \exp(-\beta(\epsilon_2 - \epsilon_1))) \right. \\ &\quad \left. + \beta \epsilon_1 + \frac{\beta(\epsilon_2 - \epsilon_1)}{\exp[\beta(\epsilon_2 - \epsilon_1)] + 1} \right] \\ &= Nk \left[\ln[1 + \exp(-\beta(\epsilon_2 - \epsilon_1))] + \frac{\beta(\epsilon_2 - \epsilon_1)}{\exp[\beta(\epsilon_2 - \epsilon_1)] + 1} \right] \end{aligned}$$

高温: $\beta \rightarrow 0$ ~~$\bar{E} = N \min$~~

$$\bar{E} \approx N \cdot \frac{\epsilon_1 + \epsilon_2}{2}, \quad S \approx Nk \ln 2$$

低温: $\beta \rightarrow \infty$

$$\bar{E} = N \min(\epsilon_1, \epsilon_2), \quad S \approx 0$$

↑
只有一个状态分布.



$$8. (7.22) \quad Z = e^{-\beta \mu B} + 1 + e^{\beta \mu B} = 1 + 2 \cosh(\beta \mu B)$$

$$n(m_j) = \frac{n}{Z} e^{+\beta \mu m_j B}$$

$$M = \frac{n}{\beta Z} \frac{\partial Z}{\partial B} = \frac{n}{\beta} \frac{\partial \ln Z}{\partial B} = \frac{n}{\beta} \cdot \frac{2 \sinh(\beta \mu B) \cdot \mu B}{1 + 2 \cosh(\beta \mu B)}$$

$$= n \mu \cdot \frac{2 \sinh(\beta \mu B)}{1 + 2 \cosh(\beta \mu B)}$$

弱场高温: $\beta B \rightarrow 0$, $\sinh(\beta \mu B) \sim \beta \mu B$
 $\cosh(\beta \mu B) \sim 1$

$$M \approx \frac{2}{3} n \mu \cdot \beta \mu B = \frac{2}{3} \cdot \frac{n \mu^2 B}{kT}$$

强场低温: $\beta B \rightarrow \infty$, $M \approx n \mu \cdot \tanh(\beta \mu B)$
 $\approx n \mu$

9. (补充题 1)

(1) $\Omega(\epsilon^r) = 8\pi^2 I \epsilon^r$, $g(\epsilon^r) d\epsilon^r = \frac{8\pi^2 I}{h^2} d\epsilon^r$

$$Z(\beta) = \int_0^\infty g(\epsilon^r) e^{-\beta \epsilon^r} d\epsilon^r = \frac{8\pi^2 I}{h^2} \int_0^\infty e^{-\beta \epsilon^r} d\epsilon^r = \frac{8\pi^2 I}{h^2 \beta}$$

$$\bar{E}^r = -N \frac{\partial \ln Z}{\partial \beta} = \frac{N}{\beta} = NkT$$

$$C_V = \left(\frac{\partial \bar{E}^r}{\partial T} \right)_V = Nk$$

$$(2) \quad Z = \sum_{j=0}^{\infty} (2j+1) e^{-\beta E_j} = \sum_{j=0}^{\infty} (2j+1) \exp\left(-\frac{\hbar^2}{2IkT} j(j+1)\right)$$

$$= \sum_{j=0}^{\infty} (2j+1) \exp\left[-\frac{h^2}{8\pi^2 IkT} j(j+1)\right]$$

~~或 $\bar{E} = \frac{\sum_{j=0}^{\infty} \frac{h^2}{8\pi^2 I} j(j+1) (2j+1) \exp\left[-\frac{h^2}{8\pi^2 IkT} j(j+1)\right]}{Z} \cdot \frac{N}{Z}$~~



$$\bar{E} = -N \frac{\partial \ln Z}{\partial \beta}$$

$$= -\frac{N}{Z} \frac{\partial Z}{\partial \beta} = \frac{N}{\sum_{j=0}^{\infty} (2j+1) \exp\left[-\frac{\beta h^2}{8\pi^2 I} j(j+1)\right]} \sum_{j=0}^{\infty} \frac{h^2}{8\pi^2 I} j(j+1)(2j+1) \exp\left[-\frac{\beta h^2}{8\pi^2 I} j(j+1)\right]$$

(3) $T \gg \theta^r, \beta \rightarrow 0$

$$Z \approx 1 + 3 \exp\left[-\frac{\beta h^2}{8\pi^2 I} \cdot 2\right] = 1 + 3 \exp\left[-\frac{2\theta^r}{T}\right], \text{取一阶近似}$$

$$\bar{E} = -N \frac{\partial \ln Z}{\partial \beta} = -N \cdot \frac{3 \exp\left[-\frac{2\beta h^2}{8\pi^2 I}\right] \cdot \left(-\frac{2h^2}{8\pi^2 I}\right)}{1 + 3 \exp\left[-\frac{2\beta h^2}{8\pi^2 I}\right]} = Nk \frac{6 \cdot \theta^r}{\exp\left(\frac{2\theta^r}{T}\right) + 3}$$

$$\approx Nk \cdot \frac{6\theta^r}{\exp\left(\frac{2\theta^r}{T}\right)} = 6Nk \theta^r \exp\left(-\frac{2\theta^r}{T}\right)$$

$$C_v = \left(\frac{\partial \bar{E}}{\partial T}\right)_v = 6Nk \theta^r \exp\left(-\frac{2\theta^r}{T}\right) \cdot \frac{2\theta^r}{T^2} = 12Nk \left(\frac{\theta^r}{T}\right)^2 \exp\left(-\frac{2\theta^r}{T}\right)$$

(4) $T \gg \theta^r, \frac{h^2}{8\pi^2 I k T} j(j+1) \rightarrow 0$

$$Z = \sum_{j=0}^{\infty} (2j+1) \exp\left[-\frac{h^2}{8\pi^2 I k T} j(j+1)\right] = \sum_{j=0}^{\infty} \int_j^{j+1} (2j+1) \exp\left[-j(j+1) \frac{\theta^r}{T}\right] dl$$

$$\approx \sum_{j=0}^{\infty} \int_j^{j+1} (2l+1) \exp\left[-l(l+1) \frac{\theta^r}{T}\right] dl$$

$$= \int_0^{\infty} (2l+1) \exp\left[-l(l+1) \frac{\theta^r}{T}\right] dl$$

$$= \int_0^{\infty} \exp\left(-t \cdot \frac{\theta^r}{T}\right) \cdot dt$$

$$= \frac{T}{\theta^r} = \frac{kT}{h^2} \cdot 8\pi^2 I = \frac{8\pi^2 I}{\beta h^2}$$

如(1)的答案般 $C_v = Nk$



$$\begin{aligned}
 10. (1) \quad Z &= \sum_{\lambda=1}^{\infty} g_{\lambda} \exp(-\beta \epsilon_{\lambda}) = g_1 \exp(-\beta \cdot 0) + g_2 \exp(-\beta \cdot \epsilon) + g_3 \exp(-\beta \cdot 10\epsilon) \\
 &\quad + \sum_{\lambda=4}^{\infty} g_{\lambda} \exp(-\beta \epsilon_{\lambda}) \\
 &= g_1 + g_2 \exp(-\beta \epsilon) + g_3 \exp(-10\beta \epsilon) + \sum_{\lambda=4}^{\infty} g_{\lambda} \exp(-\beta \epsilon_{\lambda})
 \end{aligned}$$

$$n_{\lambda} = \frac{N}{Z} g_{\lambda} e^{-\beta \epsilon_{\lambda}} \quad (\lambda=1, 2, \dots)$$

$$n_1 = \frac{N}{Z} g_1, \quad n_2 = \frac{N}{Z} g_2 \exp(-\beta \epsilon), \quad n_3 = \frac{N}{Z} g_3 \exp(-10\beta \epsilon)$$

由题意知, $\epsilon_l > \max\{\epsilon_1, \epsilon_2, \epsilon_3\} = 10\epsilon \quad (l \geq 4)$

当 $\beta \epsilon \gg 1$ 时, $\frac{n_l}{n_2} \ll 1 \quad (l \geq 3)$

$$\beta \epsilon \gg 1 \Leftrightarrow \frac{\epsilon}{kT} \gg 1 \Leftrightarrow T \ll \frac{\epsilon}{k} = \theta^T$$

$$(2) \quad \bar{E}_{\lambda} = n_{\lambda} \epsilon_{\lambda}$$

$$\bar{E}_1 = n_1 \epsilon_1 = 0, \quad \bar{E}_2 = n_2 \epsilon_2 = \frac{N}{Z} g_2 \epsilon \exp\left(-\frac{\epsilon}{kT}\right), \quad \bar{E}_3 = \frac{N}{Z} g_3 \cdot 10\epsilon \exp\left(-\frac{10\epsilon}{kT}\right)$$

$$\bar{E}_l = \frac{N}{Z} g_l \epsilon_l \exp\left(-\frac{\epsilon_l}{kT}\right)$$

$$\bar{E} = \sum_{\lambda=1}^{\infty} \bar{E}_{\lambda} = \frac{N}{Z} \left[g_2 \epsilon \exp\left(-\frac{\epsilon}{kT}\right) + 10 g_3 \epsilon \exp\left(-\frac{10\epsilon}{kT}\right) + \sum_{\lambda=4}^{\infty} g_{\lambda} \epsilon_{\lambda} \exp\left(-\frac{\epsilon_{\lambda}}{kT}\right) \right]$$

$$(3) \quad C_{v,l} = \left(\frac{\partial \bar{E}_{\lambda}}{\partial T} \right)_V = N g_l \epsilon_l \frac{\frac{\partial}{\partial T} \left(\exp\left(-\frac{\epsilon_{\lambda}}{kT}\right) \right) \cdot Z - \exp\left(-\frac{\epsilon_l}{kT}\right) \cdot \frac{\partial Z}{\partial T}}{Z^2}$$

$$= \frac{N g_l \epsilon_l}{Z} \left[\exp\left(-\frac{\epsilon_l}{kT}\right) \cdot \frac{\epsilon_l}{kT^2} - \exp\left(-\frac{\epsilon_l}{kT}\right) \cdot \frac{\partial \ln Z}{\partial \beta} \cdot \frac{\partial \beta}{\partial T} \right]$$

$$= \frac{g_l \epsilon_l}{Z} \left[N \exp\left(-\frac{\epsilon_l}{kT}\right) \frac{\epsilon_l}{kT^2} + \bar{E} \exp\left(-\frac{\epsilon_l}{kT}\right) \cdot \left(-\frac{1}{kT^2}\right) \right]$$

$$= \frac{g_l \epsilon_l}{Z k T^2} \exp\left(-\frac{\epsilon_l}{kT}\right) [N \epsilon_l - \bar{E}]$$

$$= \frac{1}{N k T^2} \bar{E}_l [N \epsilon_l - \bar{E}]$$

$$C_v = \sum_{l=1}^{\infty} C_{v,l} = \frac{1}{N k T^2} \sum_{l=1}^{\infty} (N \bar{E}_l \epsilon_l - \bar{E}_l \bar{E}) = \frac{1}{k T^2} \sum_{l=1}^{\infty} \bar{E}_l \epsilon_l - \frac{\bar{E}^2}{N k T^2}$$



當 $T \leq T^* = \frac{\epsilon}{k}$ 時, $Z \approx g_1 + g_2 \exp(-\beta \epsilon)$, $n_i \approx 0$ ($i \geq 3$)

$$\bar{E} \approx \bar{E}_1 + \bar{E}_2 = \bar{E}_2 = \frac{N}{g_1 + g_2 \exp(-\beta \epsilon)} \cdot g_2 \epsilon \exp(-\beta \epsilon)$$

$$\bar{E}_1 = 0$$

$$C_v \approx C_{v,1} + C_{v,2} = C_{v,2} = \frac{1}{NkT} \bar{E}_2 [N \epsilon_2 - \bar{E}]$$

$$= \frac{1}{NkT} \cdot \frac{N}{Z} g_2 \epsilon \exp\left(-\frac{\epsilon}{kT}\right) \left[N \epsilon - \frac{N}{Z} g_2 \epsilon \exp\left(-\frac{\epsilon}{kT}\right) \right]$$

$$= \frac{N g_2 \epsilon^2}{Z k T^2} \exp\left(-\frac{\epsilon}{kT}\right) \left[1 - \frac{g_2}{Z} \exp\left(-\frac{\epsilon}{kT}\right) \right]$$

$$= \frac{g_2}{g_1 + g_2 \exp\left(-\frac{\epsilon}{kT}\right)} \cdot N \cdot \frac{\epsilon^2}{k T^2} \exp\left(-\frac{\epsilon}{kT}\right) \left[1 - \frac{g_2}{g_1 + g_2 \exp\left(-\frac{\epsilon}{kT}\right)} \exp\left(-\frac{\epsilon}{kT}\right) \right]$$

$$= \frac{g_2}{g_1 + g_2 \exp\left(-\frac{\epsilon}{kT}\right)} \cdot N \cdot \frac{\epsilon^2}{k T^2} \exp\left(-\frac{\epsilon}{kT}\right) \cdot \frac{g_1}{g_1 + g_2 \exp\left(-\frac{\epsilon}{kT}\right)}$$

$$= \frac{N \cdot \frac{\epsilon^2}{k T^2}}{\frac{1}{g_1} + \frac{1}{\frac{g_2 \exp\left(-\frac{\epsilon}{kT}\right)}{Nk \cdot \left(\frac{T^*}{T}\right)^2}}}$$

$$= \frac{\frac{1}{g_1} + \frac{1}{\frac{g_2 \exp\left(-\frac{\epsilon}{kT}\right)}{Nk \cdot \left(\frac{T^*}{T}\right)^2}}}{\frac{1}{g_1} + \frac{1}{g_2 \exp\left(-\frac{T^*}{T}\right)}} \approx g_2 N k \left(\frac{T^*}{T}\right)^2 \exp\left(-\frac{T^*}{T}\right)$$

